

# Quadratics in factored form



1

a

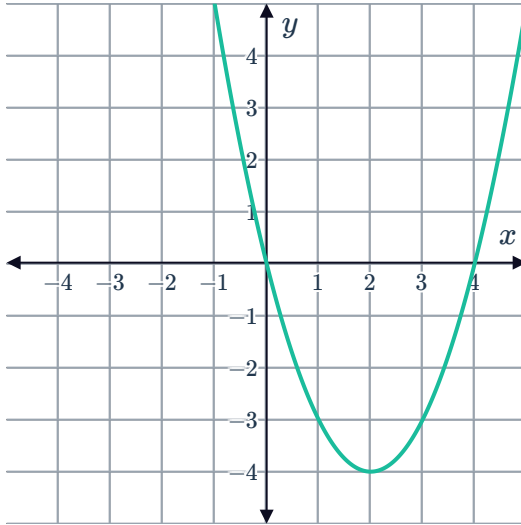
i  $y = 0$

ii  $x = 0, x = 4$

iii  $x = 2$

iv  $(2, -4)$

v



b

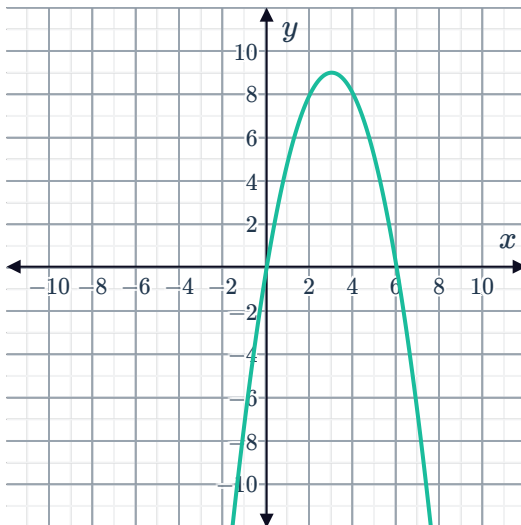
i  $y = 0$

ii  $x = 0, x = 6$

iii  $x = 3$

iv  $(3, 9)$

v

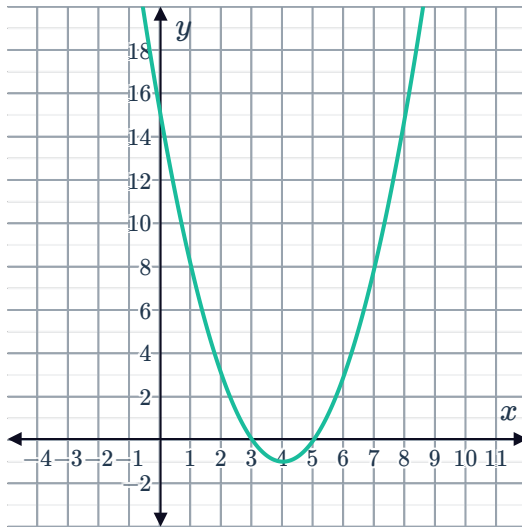


c

i  $y = 15$

iii  $x = 4$

v



ii  $x = 5, x = 3$

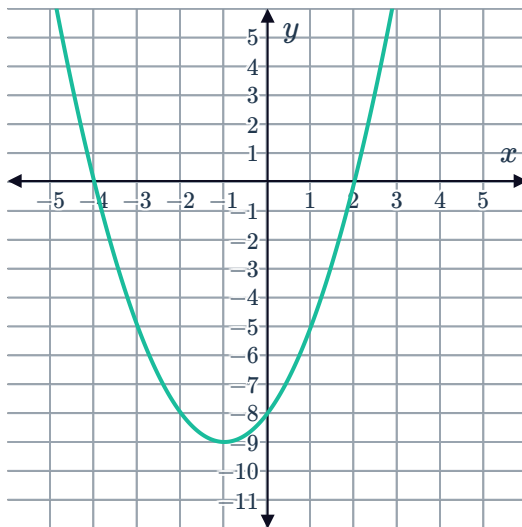
iv  $(4, -1)$

d

i  $y = -8$

iii  $x = -1$

v



ii  $x = -4, x = 2$

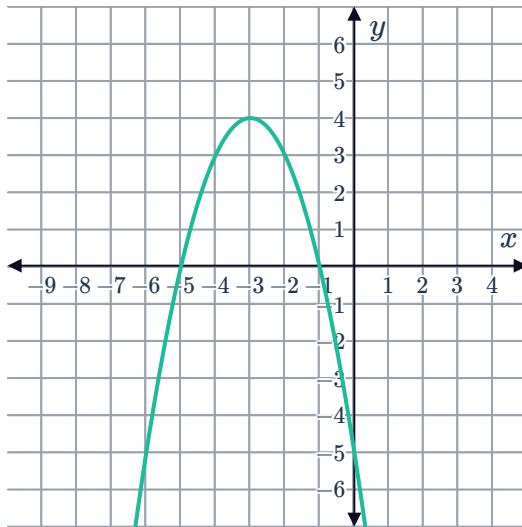
iv  $(-1, -9)$

e

i  $y = -5$

iii  $x = 3$

v



ii  $x = 1, x = 5$

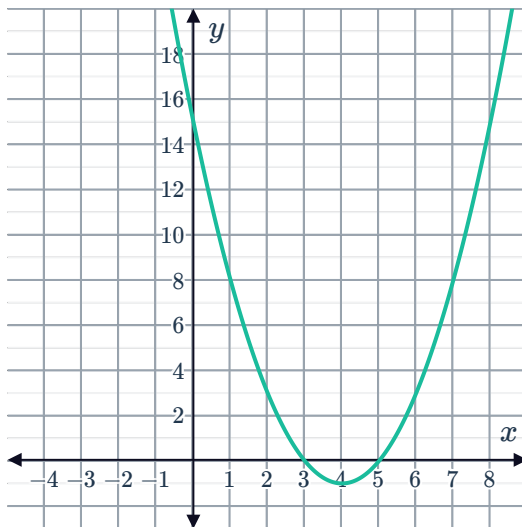
iv  $(3, 4)$

f

i  $y = 15$

iii  $x = 4$

v



ii  $x = 5, x = 3$

iv  $(4, -1)$

2



a

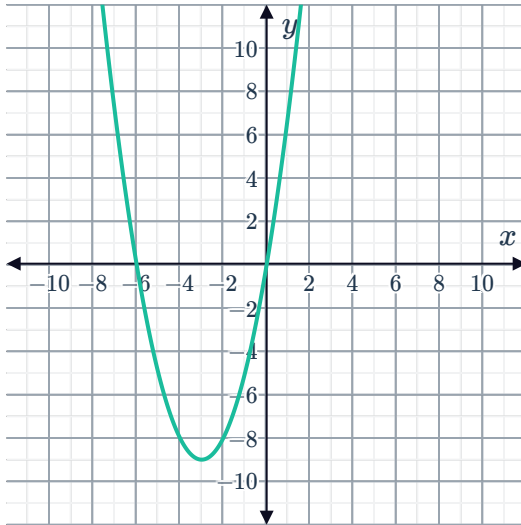
i 0

iii

|     |    |    |    |    |    |
|-----|----|----|----|----|----|
| $x$ | -5 | -4 | -3 | -2 | -1 |
| $y$ | -5 | -8 | -9 | -8 | -5 |

ii  $x = 0, x = -6$ iv  $(-3, -9)$ 

v



b

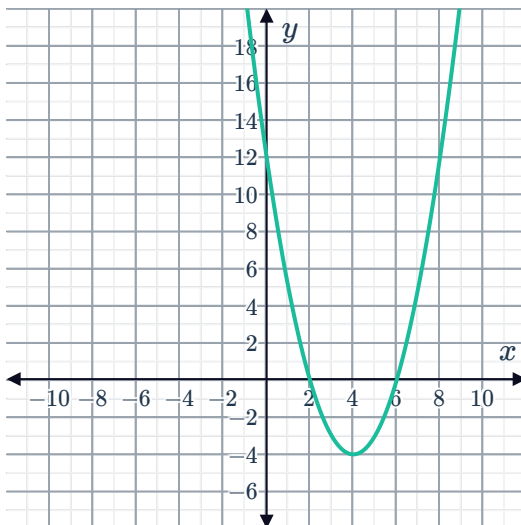
i 12

iii

|     |    |   |    |   |    |
|-----|----|---|----|---|----|
| $x$ | 0  | 2 | 4  | 6 | 8  |
| $y$ | 12 | 0 | -4 | 0 | 12 |

ii  $x = 2, x = 6$ iv  $(4, -4)$ 

v



c

i 8

iii

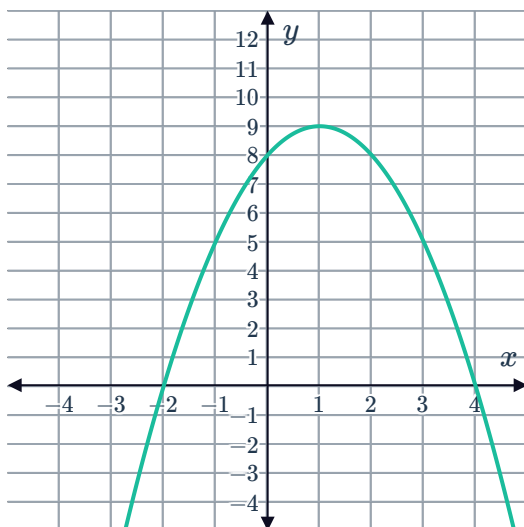
|     |    |    |    |   |    |
|-----|----|----|----|---|----|
| $x$ | -5 | -3 | -1 | 1 | 3  |
| $y$ | -7 | 5  | 9  | 5 | -7 |



ii  $x = 2, x = -4$

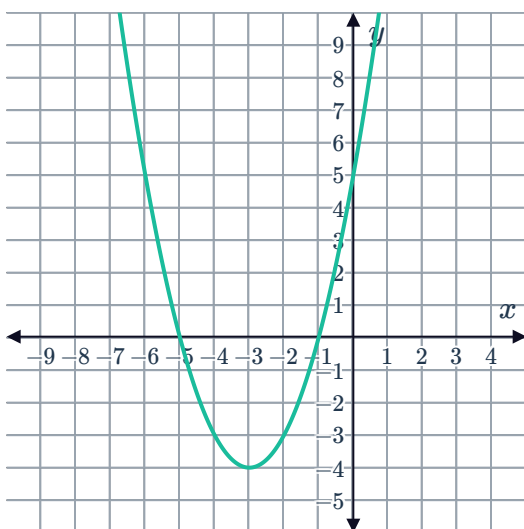
iv  $(-1, 9)$

v



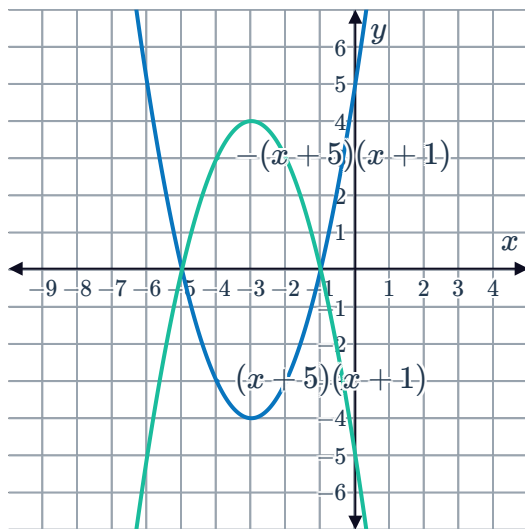
3

a



b -5

c



4

a B

b C

c D

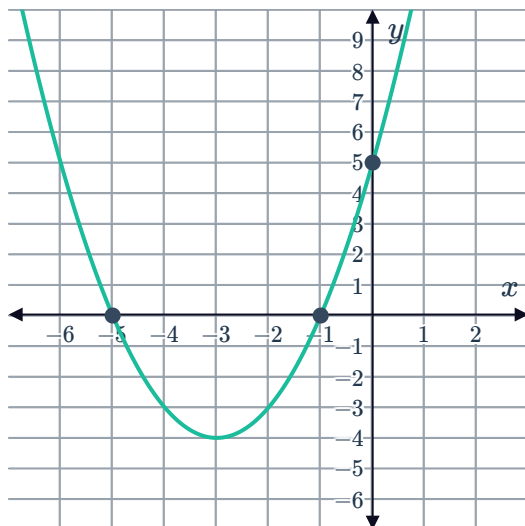
d A

5

a  $y = (x + 1)(x + 5)$

b  $y = 5$

c



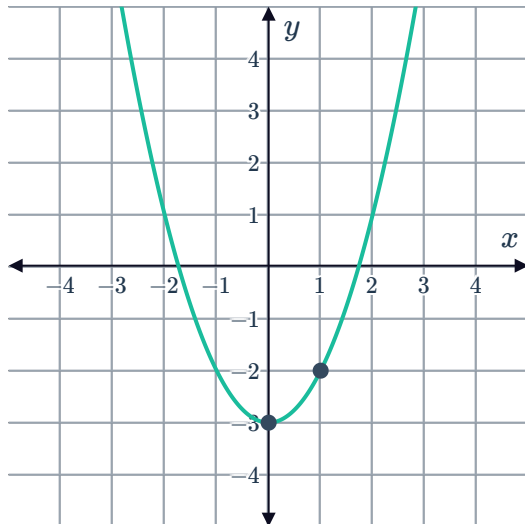
6

a  $y = x^2 - 3$



b -2

c



7

a

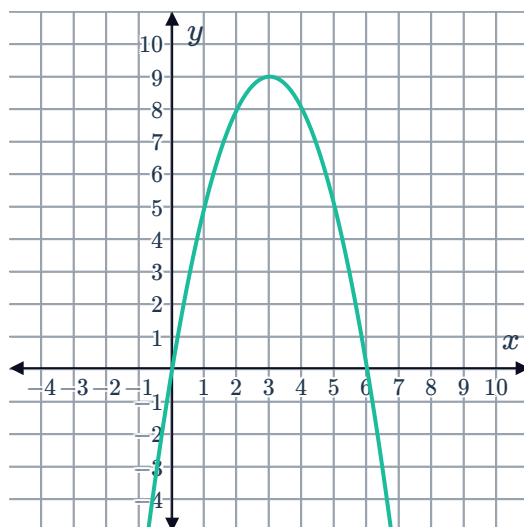
i  $x(6 - x)$

ii  $x = 0, x = 6$

iii 3

iv 9

v

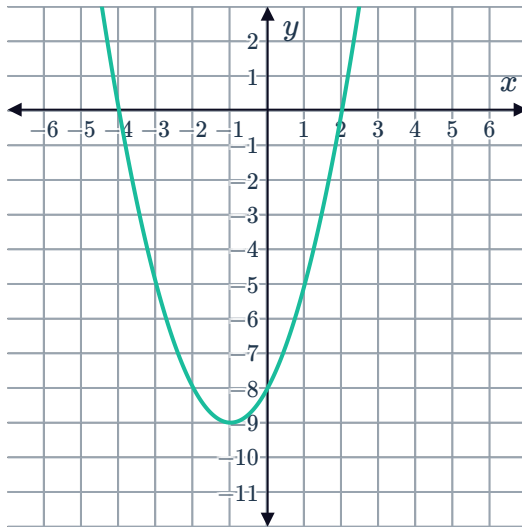


b

i  $(x - 2)(x + 4)$

iii  $-1$

v



ii  $x = 2, x = -4$

iv  $-9$

8

a  $x(4 + x)$

c  $0$

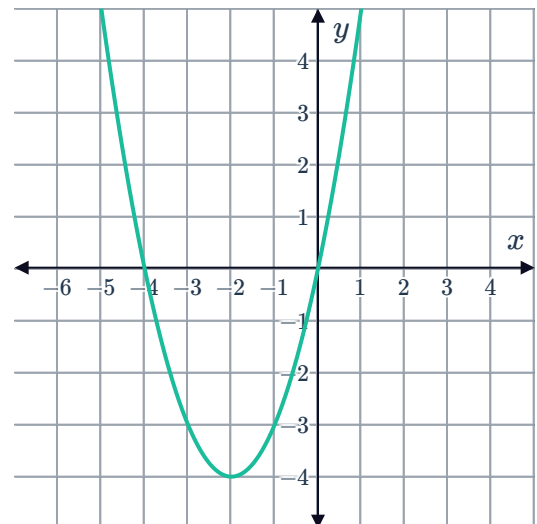
b  $x = 0, x = -4$

d

|     |      |      |      |      |     |
|-----|------|------|------|------|-----|
| $x$ | $-4$ | $-3$ | $-2$ | $-1$ | $0$ |
| $y$ | $0$  | $-3$ | $-4$ | $-3$ | $0$ |

e  $(-2, -4)$

f





9

a  $(x + 2)(x + 6)$



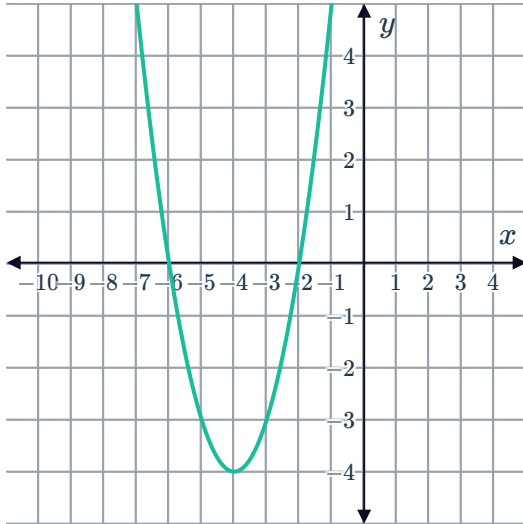
b  $x = -2, x = -6$

c

|     |    |    |    |    |    |
|-----|----|----|----|----|----|
| $x$ | -8 | -6 | -4 | -2 | 0  |
| $y$ | 12 | 0  | -4 | 0  | 12 |

d  $(-4, -4)$

e



10

a No

b Yes

11

a 8

b  $y = -(x - 2)(x + 4)$

12  $x = 0$ 

13

a  $x = -11$

b  $y = 1(x - 1)(x + 11)$

c  $(-5, -36)$

14

a  $y = x^2 - b^2$

b  $a = 0$

15

a  $t = 20$

b 100 m



## Additional questions

16  $y = a(x - x_1)(x - x_2)$

The values of  $x_1$  and  $x_2$  are the  $x$ -values of the  $x$ -intercepts of the quadratic function. The scale factor of the function is represented by  $a$ .

17 The axis of symmetry of a quadratic is the vertical line that can be drawn so each side of the quadratic function mirrors the other. This line passes through the vertex of the quadratic function.

18

a  $x = 1$  and  $4$

b  $x = 3$  and  $-7$

c  $x = 0$  and  $4$

d  $x = 1$  and  $-5$

19

a  $y = 24$

b  $y = 0$

20  $x = 1$

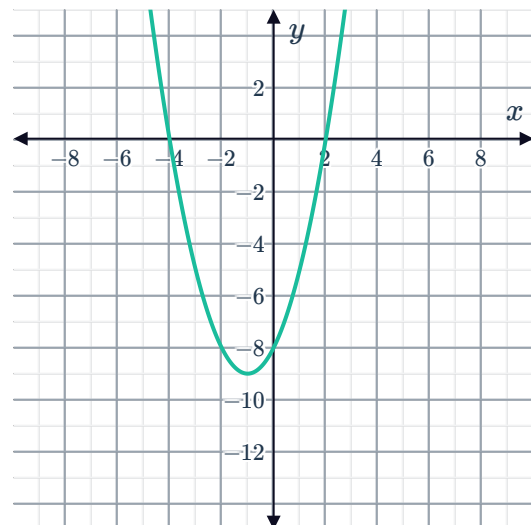
21

a i  $(2, 0)$  and  $(-4, 0)$

iii  $(-1, -9)$

ii  $x = -1$

iv

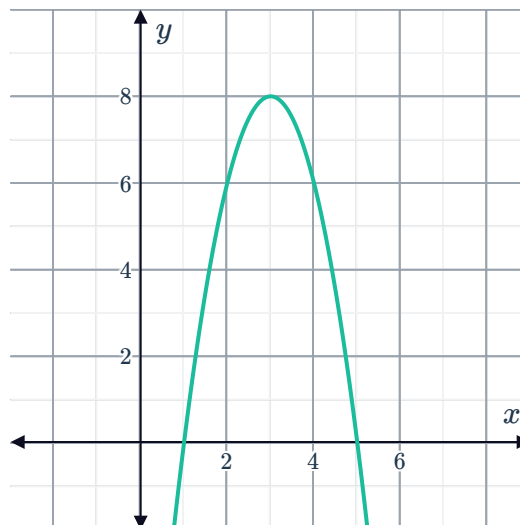


- b**
- i** (1, 0) and (5, 0)
  - iii** (3, 8)



**ii**  $x = 3$

**iv**



**22**

- a**
- i** (1, 0) and (4, 0)
  - iii**  $y = 2(x - 1)(x - 4)$
- b**
- i** (-2, 0) and (6, 0)
  - iii**  $y = -\frac{1}{3}(x - 6)(x + 2)$
- ii** (0, 8)
- ii** (0, 4)

**23**  $y = -4(x - 5)(x + 3)$

**24**  $y = \frac{1}{3}(x + 2)(x - 9)$

**25**  $y = 3(x + 4)(x + 7)$

**26**

- a** (50, 110)
- b**  $y = -\frac{11}{250}x(x - 100)$
- c** Domain:  $0 \leq x \leq 100$   
Range:  $0 \leq y \leq 110$

**27**

- a** (20, 0) and (40, 0)
- b**  $y = -\frac{1}{20}(x - 20)(x - 40)$

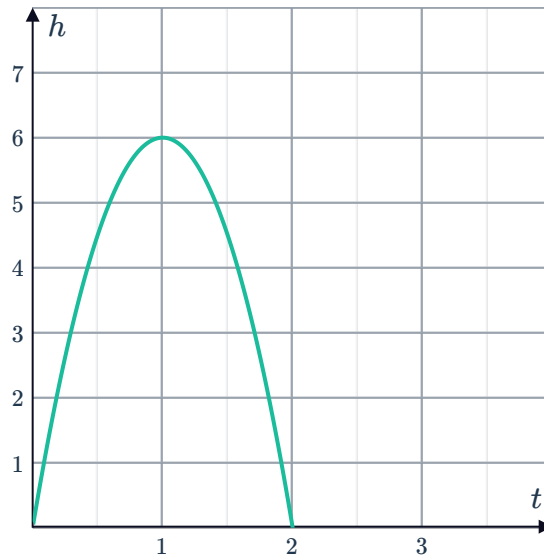


28

a The  $x$ -intercepts represent the times when the eraser is 0 ft above Wilson's hand. This will be exactly when Wilson tosses the eraser and when it returns to his hand. Since Wilson tosses the eraser at  $x = 0$  and it takes 2 seconds for it to return to his hand, the  $x$ -intercepts are  $(0, 0)$  and  $(2, 0)$ . Since one of the  $x$ -intercepts is at the origin, we have the  $y$ -intercept is also  $(0, 0)$ .

b  $(1, 6)$

c



d  $y = -6x(x - 2)$

29 Since the vertex lies on the axis of symmetry, we know that 8 is the average of 12 and  $x_0$ . So then, we must have  $x_0 = 4$ .

If we substitute the coordinates of the vertex into the function, we get:

$$15 = k(8 - 4)(8 - 12)$$

If we solve this equation we get  $k = -\frac{15}{16}$ .